



DS-471 Time Series Analysis

Project Report

Project Title: Analysis of Breast Cancer

Group No.		
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Analysis of Breast cancer

Abstract— This study investigates recovery trends among breast cancer patients subjected to varying treatment regimens. It examines the effects of demographics, cancer stages, and side effects on recovery, using data collected directly from patients. The research leverages statistical analysis and data visualization to uncover key insights and trends that can aid in optimizing treatment strategies.

I. INTRODUCTION

Breast cancer poses a significant health challenge, affecting millions of women globally. Early detection and effective treatment are critical to improving patient outcomes. This project focuses on analysing recovery trends and their relationship with patient demographics, treatment types, and associated side effects. The objective is to generate actionable insights that can enhance treatment protocols and personalize care for breast cancer patients

I. OBJECTIVE

- To explore the demographic distribution of breast cancer patients.
- To assess recovery trends across initial dose and dose readjusted.
- To identify key side effects after chemotherapy.
- To analyse the prevalence of side effects different stages.
- To identify recovery across age.

II. DATA COLLECTION

Data was collected first-hand through direct hospital visits and patient interviews. Attributes included:

Qualitative:

Demographics, cancer stage descriptions, side effect narratives.

Quantitative:

Age, recovery rates, and treatment dosages.
This comprehensive dataset captures both subjective and objective dimensions of the recovery process.

III. DATA PREPROCESSING

1. Data Collection:

- o Data was collected by visiting hospitals and interviewing patients directly.
- o Information on demographics, treatments, side effects, and recovery trends was gathered.

2. Data Cleaning:

- o Ensured data consistency and accuracy during the collection process.
- o No extensive cleaning was performed

VI. VISUALIZATION

as the data was collected first-hand without missing entries.

3. Data Analysis:

- o Conducted descriptive and ordinal statistics.
- o Visualized recovery trends using correlation heatmaps, pie charts, and boxplots.
- o Investigated side effects by stage and age groups.

4. Visualization Tools:

- o Matplotlib and Seaborn were used for data visualization.

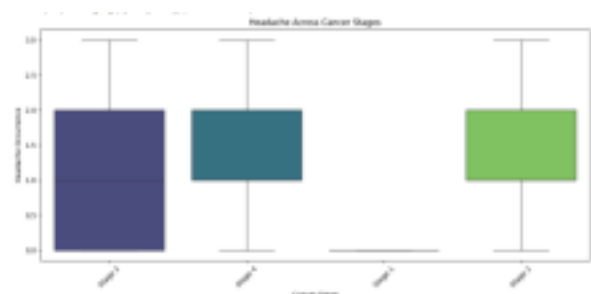
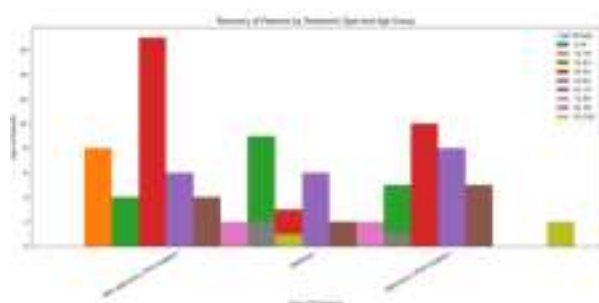
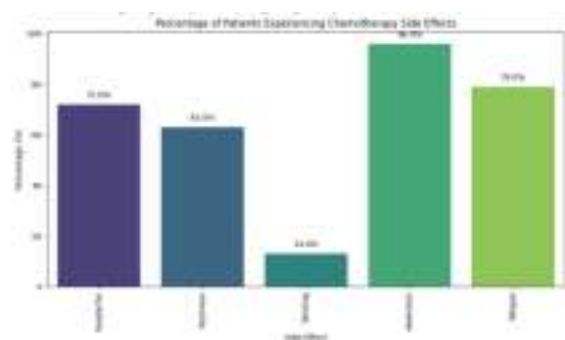
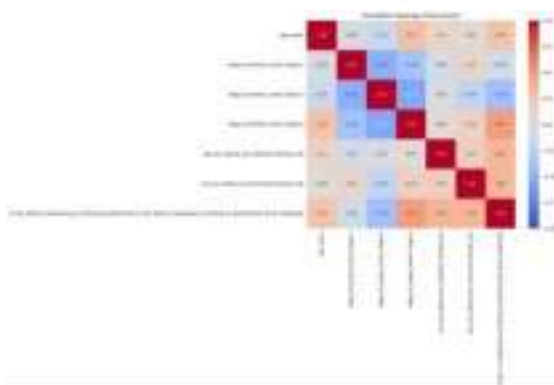
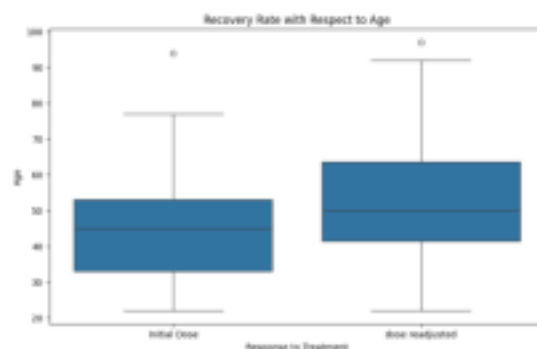
IV. DATA ANALYSIS

Descriptive Statistics:

Summarized patient demographics and recovery rates.

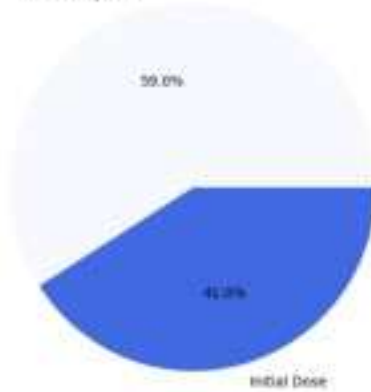
Inferential Statistics:

Correlation analyses were conducted to identify relationships between treatment factors and recovery trends.

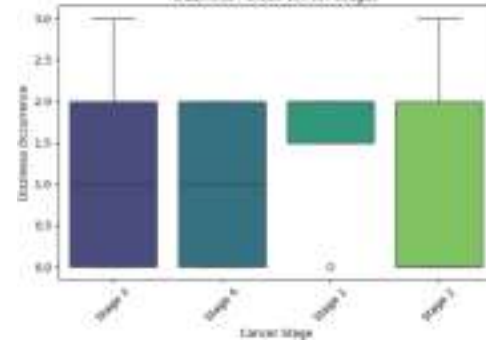


Distribution of Recovery Rate Responses

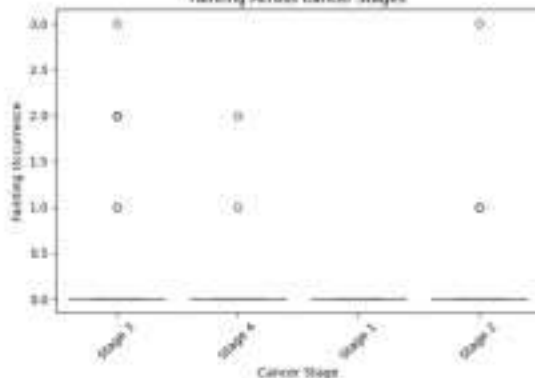
dose readjusted



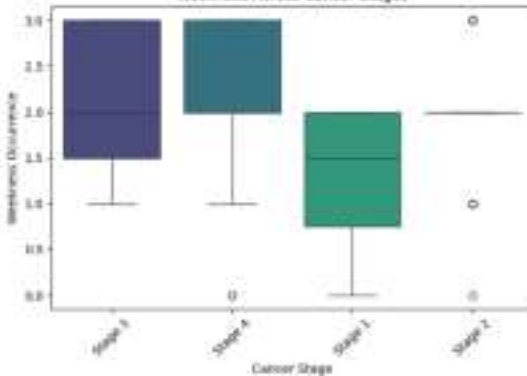
Dizziness Across Cancer Stages



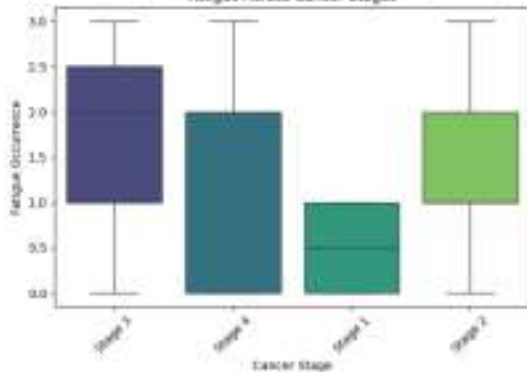
Fainting Across Cancer Stages



Weakness Across Cancer Stages



Fatigue Across Cancer Stages



areas for treatment optimization.

VIII. CONCLUSION

This project highlights significant recovery trends influenced by treatment types, demographics, and cancer stages. Side effects, stratified by age and stage, underscore the importance of personalized care strategies. Future research can build on these findings to further refine breast cancer treatment protocols and improve patient outcomes.

VII. METHODOLOGY

The project utilized descriptive and ordinal statistical methods to analyse recovery trends. Data visualization enhanced the interpretability of findings. By focusing on correlations and patterns, the study identified potential